Desheng Ma

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Cornell University
G02 Physical Sciences Building / 271 Clark Hall
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Ithaca, NY 14850

Education

2025 **Cornell University**
Ithaca, NY, USA
Ph.D., Applied Physics
Minors: Theoretical Physics and Statistics

2021 **Cornell University**
Ithaca, NY, USA
M.S., Applied Physics

2019 **Nankai University**
Tianjin, China
B.S., Physics

2016 – 2017 **Columbia University**
New York, NY, USA
Visiting student, Columbia College Visiting Student Program

Research Interests

4D-STEM, imaging theory, electron ptychography, computational imaging, ultrafast electron microscopy, electron optics, aberration correction, instrumentation, complex systems, AI for science

Research Experience

2025 – present **Brookhaven National Laboratory**
Condensed Matter Physics and Materials Science Department
Research with Dr. Yimei Zhu

- Laser-free ultrafast electron microscopy.
- Quantitative structure-factor determination using electron ptychography.

2019 – 2025 **Cornell University**

School of Applied and Engineering Physics

Ph.D. research with Prof. David Muller

- Novel imaging methods and theory in 4D-STEM:
  - 3D imaging in 4D-STEM via light-field refocusing. arXiv:2610.01057.
  - Direct ptychography with higher-order aberrations. doi:10.1093/mam/ozag008.
  - Information transfer in 4D-STEM. doi:10.1016/j.ultramic.2026.114351.
  - Information limit and dose efficiency of electron ptychography. doi:10.1093/mam/ozae044.910.
- Emittance minimization for aberration correction via machine learning and Bayesian optimization. Part I; Part II.

- Multi-objective Bayesian optimization for electron ptychography.
- In situ diffraction and 4D-STEM imaging of 2H-$TaSe_2$ CDW domains. arXiv:2603.15915.

2020 – 2022 **Cornell University**
Samuel Curtis Johnson Graduate School of Management
Research assistant with Prof. Shawn Mankad

- A Structural Model for Latent Ratings and Topics in Text (draft available).
- CP-Squared: A method for change point detection in core–periphery networks. doi:10.1016/j.eswa.2022.116660.

Summers 2017 and 2018 **Columbia University**
Department of Physics
Research intern with Prof. Sebastian Will

- New experimental setup for the laser cooling and trapping of ultracold dipolar molecules.

Work in Progress

[1] **D. Ma**, C.-H. Lee, Z. Shi, D. A. Muller, and S. E. Zeltmann. "Parallax Depth Sectioning and 3D Reconstruction in 4D-STEM". Preprint. Oct. 1, 2026. arXiv: 2610.01057 [physics.optics].

[2] A. Raji, J. Luo, M. A. Guercio, C. Huang, **D. Ma**, S. Yan, A. C. Marschilok, E. S. Takeuchi, K. J. Takeuchi, and Y. Zhu. "Microscopic observation of surface spinel formation in a Mn-rich high-entropy layered oxide tuned by Li-stoichiometry". Submitted. Oct. 2026.

[3] H. Yang, J. Kim, **D. Ma**, D. Yoon, D. G. Schlom, and D. A. Muller. "Three-dimensional imaging of oxygen dopant distribution in $Sr_2CuO_{3+\delta}$ by electron ptychography". Preprint. Aug. 20, 2026. arXiv: 2608.20199.

[4] H. Huang, J. Dai, J. Christiansen-Salameh, J. Kim, S. Kielar, **D. Ma**, N. Schinitzer, D. Ni, G. Alvarez, C. Li, C. Slebodnick, M. Medina, B. Azhar, A. Alatas, R. Cava, D. Muller, and Z. Tian. "Anomalous Thermal Conductivity Reversal Driven by a Fluctuating Charge Density Wave". Preprint. Mar. 16, 2026. arXiv: 2603.15915.

Publications

Google Scholar

[1] **D. Ma**, L. Wu, D. A. Muller, and Y. Zhu. "Advances in 4D-STEM for Quantum Materials". In: *Microscopy and Microanalysis* 32.Supplement 1, ozag053.577 (July 2026). DOI: 10.1093/mam/ozag053.577.

[2] **D. Ma**, S. E. Zeltmann, and D. A. Muller. "Why Ptychography Works Better with Aberrations". In: *Microscopy and Microanalysis* 32.Supplement 1, ozag053.053 (July 2026). DOI: 10.1093/mam/ozag053.053.

[3] G. Li, **D. Ma**, and D. A. Muller. "Direct High-Resolution Imaging of Covalent Organic Framework Defects". In: *Microscopy and Microanalysis* 32.Supplement 1, ozag053.509 (July 2026). DOI: 10.1093/mam/ozag053.509.

[4] S. E. Zeltmann, **D. Ma**, and D. A. Muller. "Improving Low-Dose Three-Dimensional Imaging with Tilt-Corrected 4D-STEM". In: *Microscopy and Microanalysis* 32.Supplement 1, ozag053.447 (July 2026). DOI: 10.1093/mam/ozag053.447.

[5] **D. Ma**, G. Li, D. A. Muller, and S. E. Zeltmann. "Information in 4D-STEM: Where it is, and How to Use it". In: *Ultramicroscopy* 283, 114351 (May 2026). DOI: 10.1016/j.ultramic.2026.114351.

[6] **D. Ma**, D. A. Muller, and S. E. Zeltmann. "Using Aberrations to Improve Dose-Efficient Tilt-corrected 4D-STEM Imaging". In: *Microscopy and Microanalysis* 32.2, ozag008 (Apr. 2026). DOI: 10.1093/mam/ozag008.

[7] G. Li, Z. Shi, **D. Ma**, J. Zhang, X. Guo, H. Zhang, and Y. Han. "Depth-Resolved Imaging of Surfaces and Interfaces in MFI-Type Zeolite Crystals via Multislice Electron Ptychography". In: *Advanced Materials*, e19613 (Mar. 18, 2026). DOI: 10.1002/adma.202519613.

[8] R.-J. Jiang, M.-X. Zhu, S.-Z. Liu, Y.-T. Chen, **D. Ma**, Y.-P. Feng, M.-J. Zou, M.-J. Han, C. H. Lei, Y.-J. Wang, Y.-L. Tang, Y.-L. Zhu, and X.-L. Ma. "Strong interplay between polar and structural topologies". In: *Nature Communications* 17, 3882 (Mar. 13, 2026). DOI: 10.1038/s41467-026-70515-y.

[9] Center for AI Safety, Scale AI, and HLE Contributors Consortium. "A benchmark of expert-level academic questions to assess AI capabilities". In: *Nature* 649 (Jan. 28, 2026), pp. 1139–1146. DOI: 10.1038/s41586-025-09962-4.

[10] U. Pratiush, A. Houston, K. Barakati, A. Raghavan, R. Bulanadi, X. Yin, S. S. Welborn, D. Yoon, Harikrishnan KP, Z. Baraissov, **D. Ma**, et al. "Mic-hackathon 2024: hackathon on machine learning for electron and scanning probe microscopy". In: *Machine Learning: Science and Technology* 6.4, 040701 (Dec. 30, 2025). DOI: 10.1088/2632-2153/ae1f5d.

[11] Z. Shi, Q. Li, H. Mishra, **D. Ma**, H. D. Abruña, and D. A. Muller. "Electron ptychography images hydrogen atom superlattices and 3D inhomogeneities in palladium hydride nanoparticles". In: *Journal of the American Chemical Society* 147.43 (Oct. 10, 2025), pp. 39020–39025. DOI: 10.1021/jacs.5c13741.

[12] **D. Ma**, S. Zeltmann, C. Zhang, et al. "Emittance Minimization for Aberration Correction I: Aberration correction of an electron microscope without knowing the aberration coefficients". In: *Ultramicroscopy* 273, 114137 (July 2025). DOI: 10.1016/j.ultramic.2025.114137.

[13] **D. Ma**, S. Zeltmann, C. Zhang, et al. "Emittance Minimization for Aberration Correction II: Physics-informed Bayesian Optimization of an Electron Microscope". In: *Ultramicroscopy* 273, 114138 (July 2025). DOI: 10.1016/j.ultramic.2025.114138.

[14] **D. Ma**, S. E. Zeltmann, C. Zhang, et al. "Emittance Minimization for Aberration Correction via Machine Learning and Bayesian Optimization". In: *Microscopy and Microanalysis* 31.Supplement 1, ozaf048.1068 (July 2025). DOI: 10.1093/mam/ozaf048.1068.

[15] Z. Shi, **D. Ma**, M. I. Bardot, W. Dichtel, and D. A. Muller. "Atomic Resolution Imaging of 2D Polymers with Tilt-corrected Bright Field (tcBF) Imaging". In: *Microscopy and Microanalysis* 31.Supplement 1, ozaf048.074 (July 2025). DOI: 10.1093/mam/ozaf048.074.

[16] S. E. Zeltmann, **D. Ma**, and D. Muller. "A non-iterative phase retrieval method for defocused, coarsely scanned 4D-STEM data". In: *Microscopy and Microanalysis* 31.Supplement 1, ozaf048.034 (July 2025). DOI: 10.1093/mam/ozaf048.034.

[17] W.-R. Geng, X. Guo, Y.-L. Zhu, **D. Ma**, Y.-L. Tang, Y.-J. Wang, Y. Wu, Z. Hong, and X.-L. Ma. "Observation of multi-order polar radial vortices and their topological transition". In: *Nature Communications* 16, 2804 (Mar. 21, 2025). DOI: 10.1038/s41467-025-58008-w.

[18] C.-H. Lee, S. Zeltmann, D. Yoon, **D. Ma**, and D. Muller. "PtyRAD: A High-performance and Flexible Ptychographic Reconstruction Framework with Automatic Differentiation". In: *Microscopy and Microanalysis* 31.4, ozaf070 (2025). DOI: 10.1093/mam/ozaf070.

[19] **D. Ma**. "Decoding Aberrations and Information in Phase Space for Electron Microscopy". PhD thesis. Cornell University, 2025. DOI: 10.7298/zj2g-y804.

[20] **D. Ma** and D. Muller. "Information Limit and Dose Efficiency of Electron Ptychography". In: *Microscopy and Microanalysis* 30.Supplement 1, ozae044.910 (July 2024). DOI: 10.1093/mam/ozae044.910.

[21] **D. Ma**, C. Zhang, Y. T. Shao, et al. "Physics-informed Bayesian Optimization of an Electron Microscope". In: *Microscopy and Microanalysis* 29.Supplement 1 (July 2023), pp. 1875–1877. DOI: 10.1093/micmic/ozad067.968.

[22] **D. Ma** and S. Mankad. "CP-Squared: A method for change point detection in core–periphery networks". In: *Expert Systems with Applications* 196, 116660 (June 2022). DOI: 10.1016/j.eswa.2022.116660.

[23] Y. Feng, Y. Tang, **D. Ma**, et al. "Thickness-dependent evolution of piezoresponses and stripe 90 domains in (101)-oriented ferroelectric PbTiO3 thin films". In: *ACS Applied Materials & Interfaces* 10 (2018), pp. 24627–24637.

[24] M. Han, Y. Wang, **D. Ma**, et al. "Coexistence of rhombohedral and orthorhombic phases in ultrathin BiFeO3 films driven by interfacial oxygen octahedral coupling". In: *Acta Materialia* 145 (2018), pp. 220–226.

[25] S. Zhang, X. Guo, Y. Tang, **D. Ma**, et al. "Polarization rotation in ultrathin ferroelectrics tailored by interfacial oxygen octahedral coupling". In: *ACS Nano* 12 (2018), pp. 3681–3688.

Conferences and Workshops

Aug. 2026 — **P04.1: Advances in 4D-STEM for Quantum Materials** (invited talk)
Microscopy & Microanalysis 2026 Meeting, August 2–6, Milwaukee, WI, USA

Aug. 2026 — **A02.P1: Why Ptychography Works Better with Aberrations** (poster #97)
Microscopy & Microanalysis 2026 Meeting, August 2–6, Milwaukee, WI, USA

May 2026 — **CH04.07.07: Dose-Efficient Tilt-Corrected 4D-STEM Imaging Using Aberrations**
2026 MRS Spring Meeting & Exhibit, April 26–May 1, Honolulu, HI, USA

July 2025 — **C06.3: Emittance Minimization for Aberration Correction via Machine Learning and Bayesian Optimization**
Microscopy & Microanalysis 2025 Meeting, July 27–31, Salt Lake City, UT, USA

Mar. 2025 — **MAR-S39: Physics-Informed Bayesian Optimization of an Electron Microscope**
APS March Meeting 2025, March 15–22, Los Angeles, CA, USA

July 2024 — **C01.3: Information Limit and Dose Efficiency of Electron Ptychography**
Microscopy & Microanalysis 2024 Meeting, July 28–August 1, Cleveland, OH, USA

July 2023 — **C01.2: Physics-Informed Bayesian Optimization of an Electron Microscope**
Microscopy & Microanalysis 2023 Meeting, July 23–27, Minneapolis, MN, USA

Mar. 2023 — **A45.00007: Physics-Informed Bayesian Optimization of an Electron Microscope**
APS (American Physical Society) March Meeting 2023, March 5–10, Las Vegas, NV, USA

Dec. 2022 — **Physics-informed Bayesian Optimization of an Electron Microscope** (poster)
Machine Learning and the Physical Sciences Workshop, NeurIPS 2022, December 3, New Orleans, LA, USA

Nov. 2022 — **Physics-Informed Bayesian Optimization of an Electron Microscope**
MIT.nano workshop: Automation, AI and Machine Learning in Electron Microscopy: Current Trends and Future Developments, Boston, MA, USA

Nov. 2022 — **Machine Learning for Beam Emittance Measurement and Aberration Correction of an Electron Microscope**
3rd ICFA Beam Dynamics Mini-Workshop on Machine Learning Applications for Particle Accelerators, Chicago, IL, USA

Oct. 2022 — **SMRT: A Structural Model of Latent Ratings and Topics in Text**

- INFORMS Workshop on Data Science 2022, Indianapolis, IN, USA. Awarded the INFORMS Workshop on Data Science 2022 Student Scholarship.
- Conference on Information Systems and Technology (CIST) 2022, Indianapolis, IN, USA.
- 17th INFORMS Workshop on Data Mining and Decision Analytics (DMDA), Indianapolis, IN, USA. Best Paper Finalist.

June 2022 — **Automation of Electron Microscope Aberration Corrector Tuning***
The Center for Bright Beams Annual Meeting 2022, Los Angeles, CA, USA

June 2022 **SMRT: A Structural Model of Latent Ratings and Topics in Text** (oral presentation)
2022 Symposium on Data Science & Statistics (SDSS), Pittsburgh, PA, USA

Jan. 2021 **Change Detection in Core-Periphery Networks: A Case Study on Detecting Financial Crises in the Interbank Market** (oral presentation)
Cornell 2021 Day of Data, Ithaca, NY, USA

Mar. 2018 **A09.00003: Evolution of a1/a2 Domains in Strained Ferroelectric Thin Films**
APS (American Physical Society) March Meeting 2018, March 5–9, Los Angeles, CA, USA

Teaching Experience

Cornell University

Spring 2022	**AEP 4340**	Fluid and Continuum Mechanics
Fall 2021	**AEM 4280**	Valuation of Capital Investment
Fall 2021	**AEP 3330**	Mechanics of Particles and Solid Bodies
Summer 2021	**NBA 6550**	Introduction to SQL
Summer 2021	**NCC 5060**	Managerial Finance
Spring 2020	**AEP 3200**	Mathematical Physics
Fall 2019	**AEP 1100**	Lasers and Photonics

Education and Outreach

2026 **MicroscopeCraft: Educational Game Development**
Contributed to a free, open-source browser game introducing students and the public to microscopy through interactive instrument building and exploration.

- Learning activities connect optical and electron microscopy with instrument alignment, aberration correction, and ptychographic reconstruction.
- Publicly available game and demonstration video support self-directed exploration of imaging physics.

Game | Source code | Video demonstration

Honors and Awards

2022 **3rd ICFA Beam Dynamics Mini-Workshop Student Scholarship**
The Center for Bright Beams

2022 **INFORMS Workshop on Data Science 2022 Student Scholarship**
INFORMS

2018 **Travel Award, APS March Meeting 2018**
Nankai University

2017 **Boling Class Fellowship**
Nankai University

2016 **First Prize, Nankai Physics Tournament**
Nankai University

2015 **Boling Class Freshman Fellowship**
Nankai University

Editorial and Peer Review Service

Editorial boards: Early Career Editorial Board member, *Materials Letters* and *Materials Letters: X*.

Journal reviewer: Physical Review Letters; Acta Materialia; npj Computational Materials; Physical Review B; Carbon Energy; Physical Review Materials; IEEE Transactions on Computational Imaging; Microscopy and Microanalysis; Energy & Fuels; Optics & Laser Technology; Materials Characterization; Scientific Reports; Materials Letters; Microstructures; Data Science in Science; AIMS Energy; International Journal of Technology and Information Studies (IJTIS).

Conference reviewer: International Symposium on Computational Sensing (ISCS), 2025.

Professional References

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Prof. David Muller
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Argonne National Laboratory, Lemont, IL, USA
yjiang@anl.gov